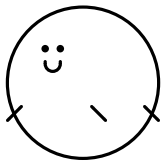


A frog sits in a bog. What does it sound like?

A frog sits in a bog. What does it sound like?



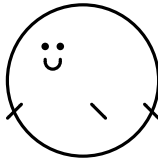
$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$
$$\rho \frac{d\mathbf{v}}{dt} = -\nabla P$$

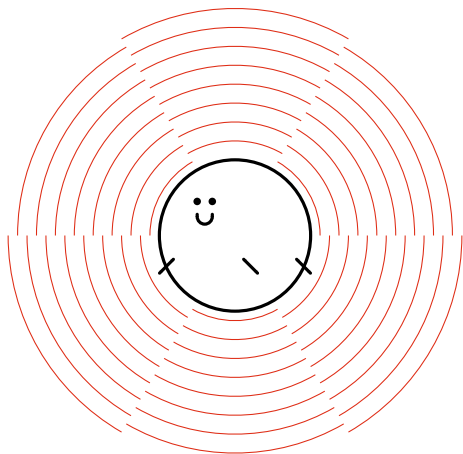
$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$
$$\rho \left[\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right] = -\nabla P$$

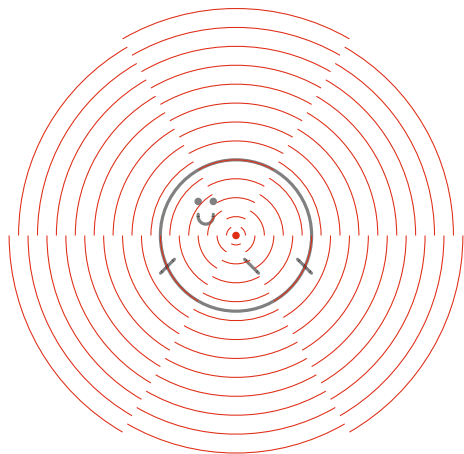
$$\mathbf{v} = \nabla \phi$$

$$P = P_0 - \rho_0 \frac{\partial \phi}{\partial t}$$

$$-\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} + \nabla^2 \phi = 0$$





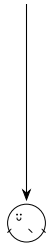


$$-\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} + \nabla^2 \phi = q$$

$$\phi(t, \mathbf{r}) = -\frac{1}{4\pi} \int \frac{q(t_{\text{ret}}, \mathbf{r}')}{\|\mathbf{r} - \mathbf{r}'\|} d^3r'$$



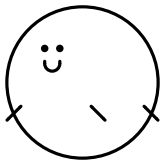
frog

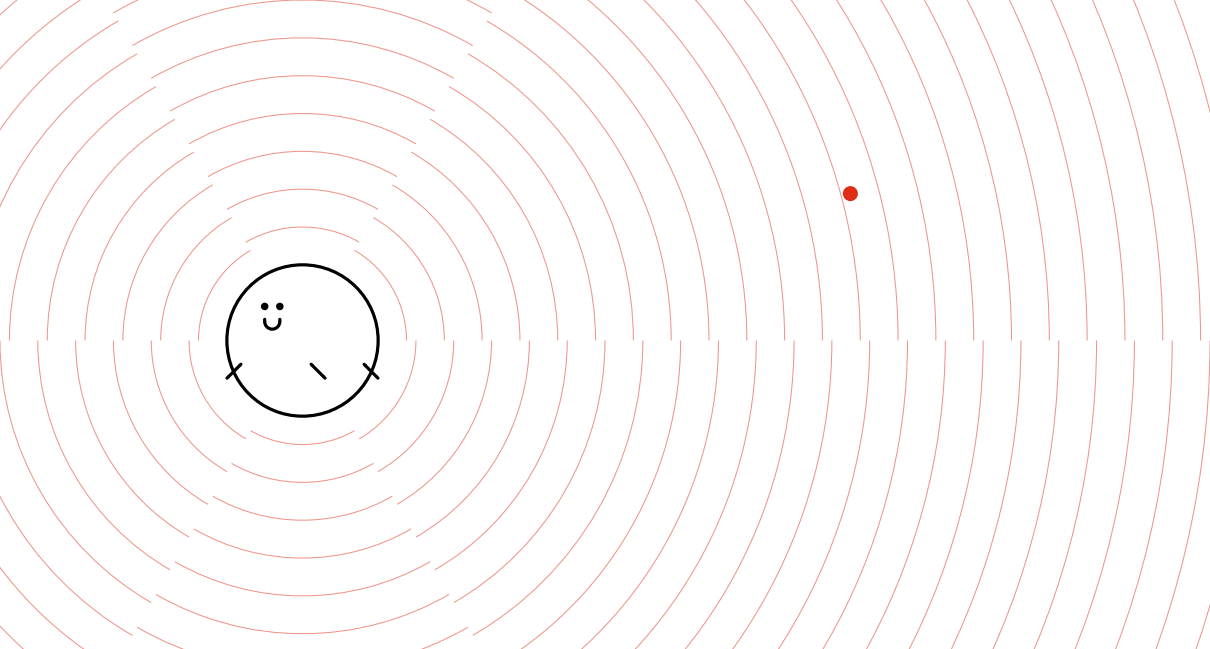


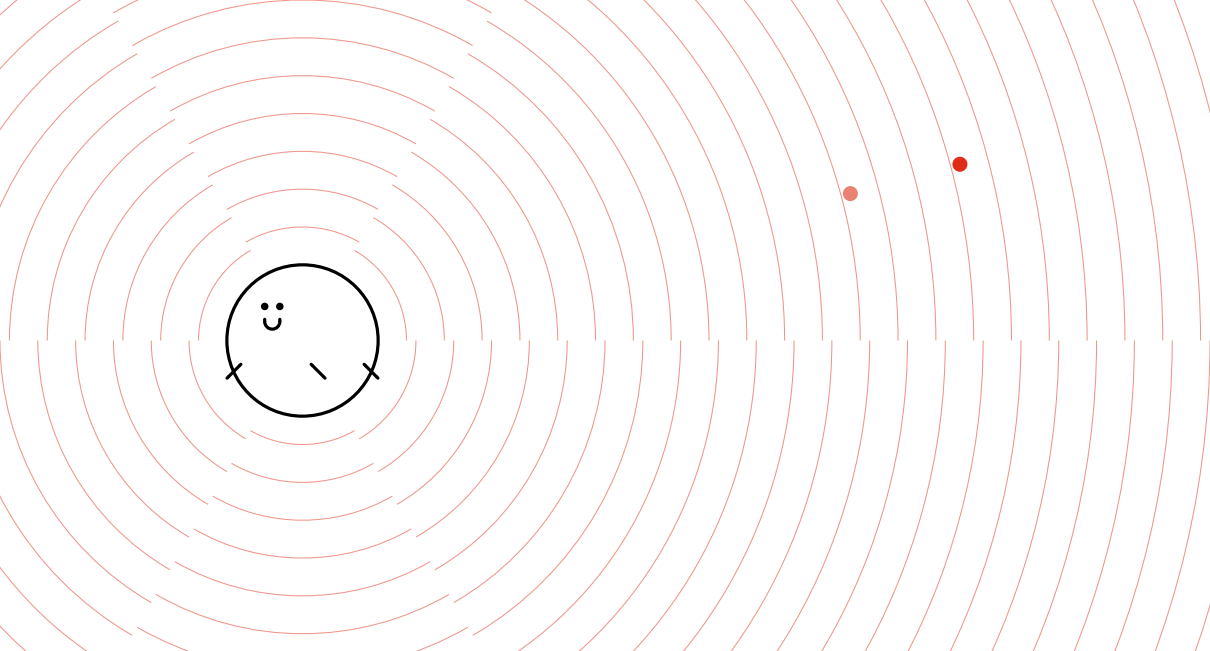
$$\phi(t, \mathbf{r}) = -\frac{1}{4\pi r} \int q\left(t - \frac{r}{c} + \frac{\hat{\mathbf{r}} \cdot \mathbf{r}'}{c}, \mathbf{r}'\right) d^3 r' + \mathcal{O}\left(\frac{1}{r^2}\right)$$

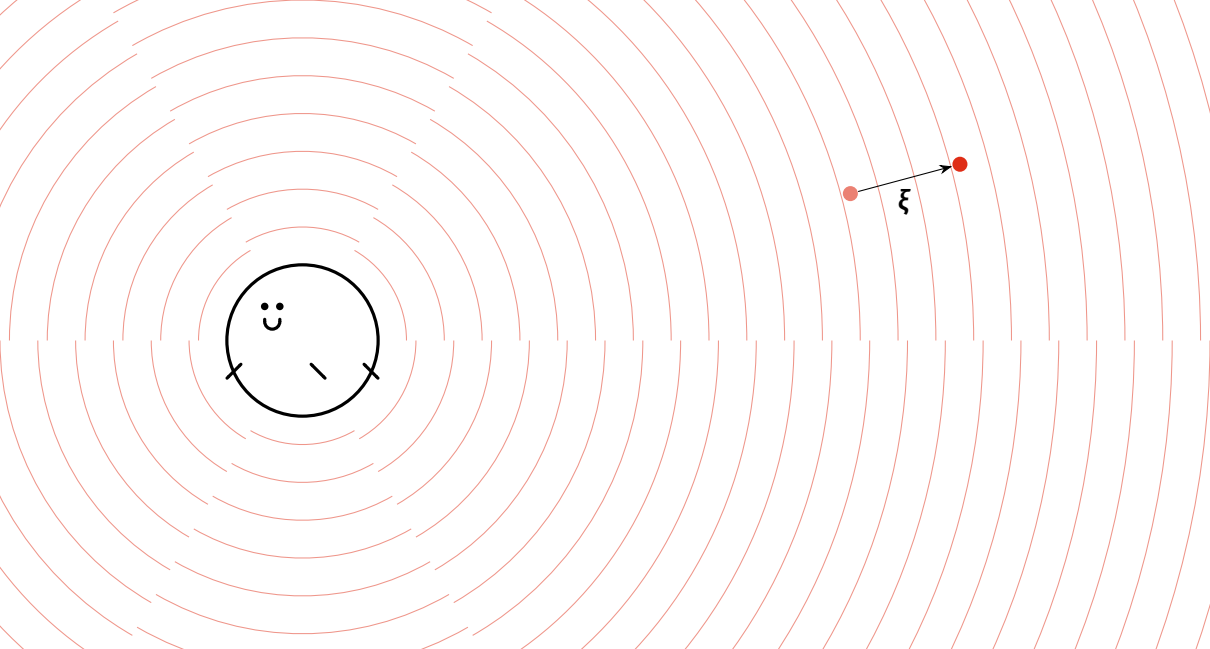
$$\phi(t, \mathbf{r}) = -\frac{Q\left(t - \frac{r}{c}, \hat{\mathbf{r}}\right)}{4\pi r} + \mathcal{O}\left(\frac{1}{r^2}\right)$$

$$\mathbf{v}(t, \mathbf{r}) = \frac{\dot{Q}\left(t - \frac{r}{c}, \hat{\mathbf{r}}\right) \hat{\mathbf{r}}}{4\pi c r} + \mathcal{O}\left(\frac{1}{r^2}\right)$$





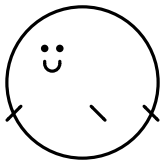


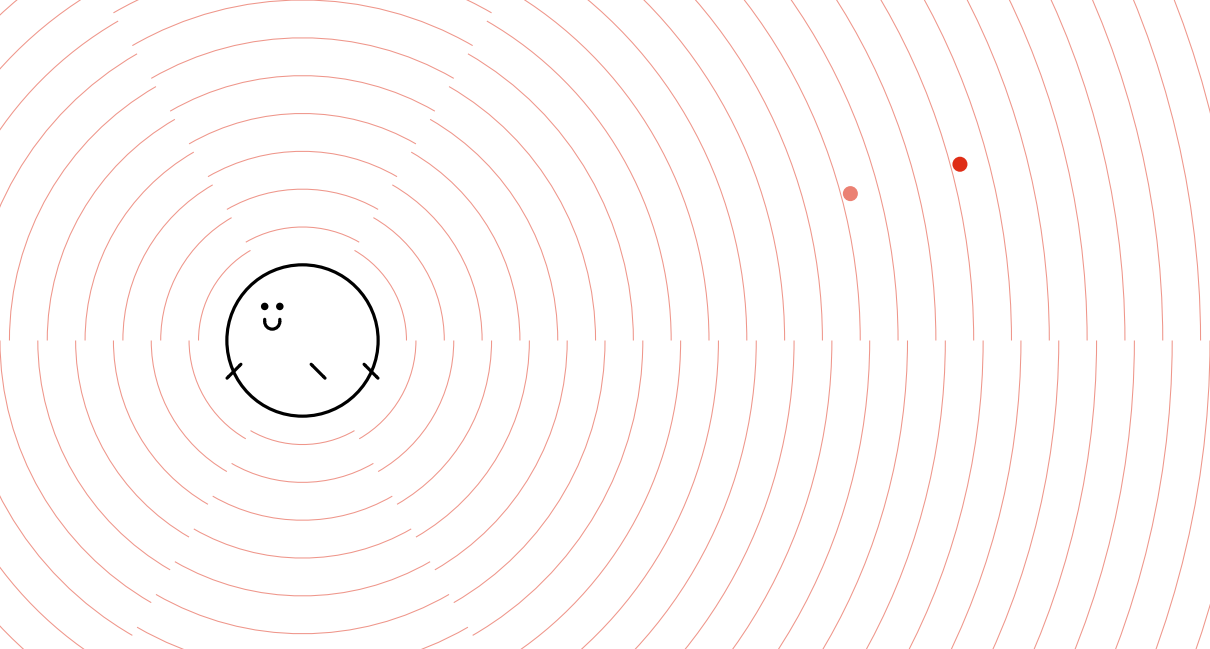


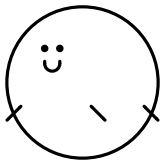
$$\frac{\partial \xi}{\partial t} = \mathbf{v} + \dots$$

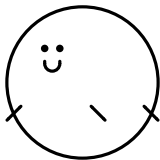
$$\frac{\partial \xi}{\partial t} = \frac{\dot{Q}\left(t - \frac{r}{c}, \hat{\mathbf{r}}\right) \hat{\mathbf{r}}}{4\pi c r} + \dots$$

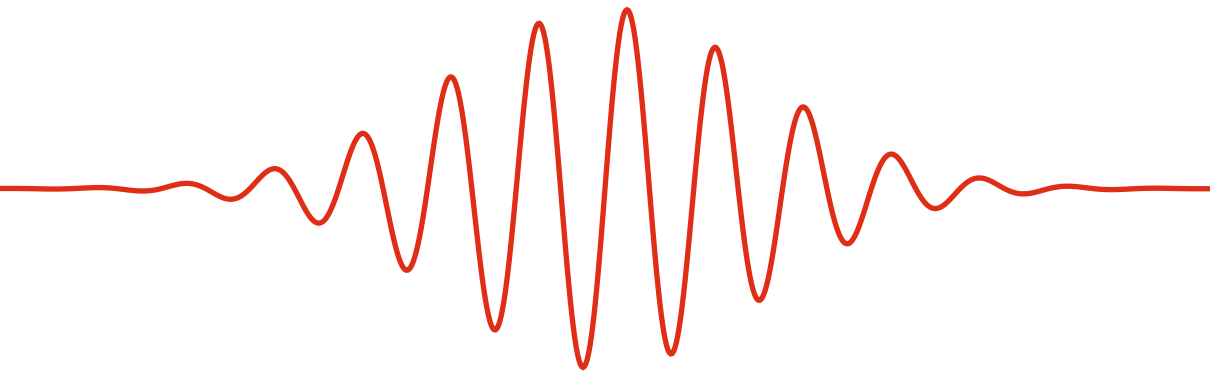
$$\Delta\xi = \frac{\Delta Q\left(t - \frac{r}{c}, \hat{\mathbf{r}}\right)\hat{\mathbf{r}}}{4\pi cr} + \dots$$

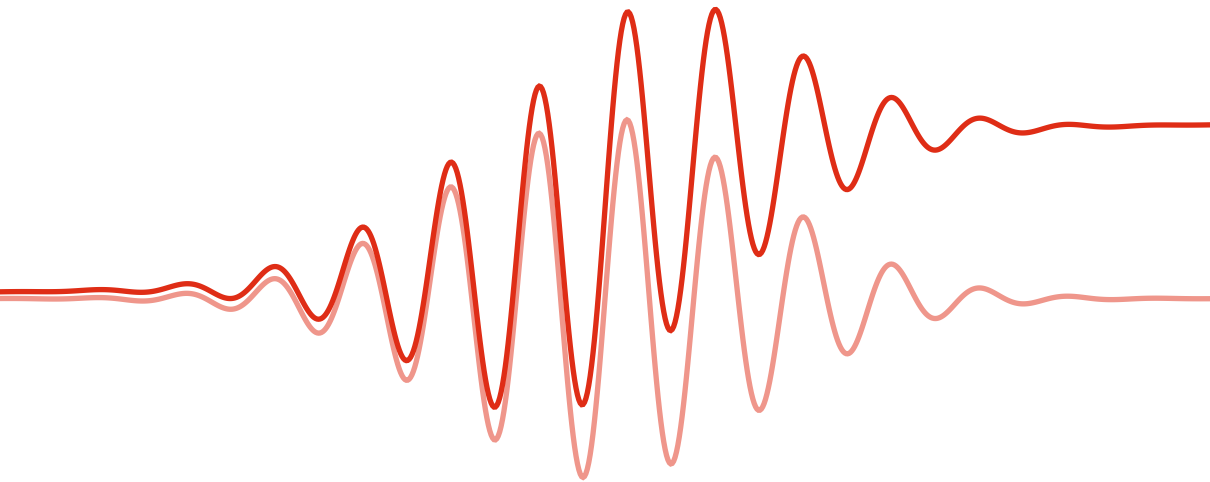












Why talk about frogs in a conference about quantum gravity, early universe, spacetime, and theoretical frontiers?

Can frogs teach us something about gravity?

Níckolas de Aguiar Alves
(he/him)

PhD candidate at Federal University of ABC, Brazil
Visiting at Scuola Normale Superiore, Italy

23 September 2026

- Acoustics: **Níck** & Landulfo 2025

- Acoustics: **Nick** & Landulfo 2025
- Electromagnetism: Bieri & Garfinkle 2013

- **Acoustics:** **Níck** & Landulfo 2025
- **Electromagnetism:** Bieri & Garfinkle 2013
- **Gravity:** Zel'dovich & Polnarev 1974

- **Acoustics:** **Níck** & Landulfo 2025
- **Electromagnetism:** Bieri & Garfinkle 2013
- **Gravity:** Zel'dovich & Polnarev 1974

None have been experimentally confirmed. **Yet.**

Memory is wrong?

Memory is wrong?

- Then we fundamentally misunderstood waves

Memory is wrong?

→ Then we fundamentally
misunderstood waves

Memory is right?

Memory is wrong?

- Then we fundamentally misunderstood waves

Memory is right?

- Then there are other implications to fundamental physics

$$-\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} + \nabla^2 \phi = 0$$

$$d\star d\phi = 0$$

$$d\star d\phi = 0$$

$$dd\phi = 0$$

$$d \star d\phi = 0$$

$$d \star \star d\phi = 0$$

$$d \star d\phi = 0$$

$$d \star \star d\phi = 0$$

$$dH = 0$$

$$d\star H = 0$$

$$dH = 0$$

$$d\star H = 0$$

Linear Acoustics

$$dH = 0$$

$$d\star H = 0$$

H is a three-form

Linear Acoustics

$$dH = 0$$

$$d\star H = 0$$

H is a three-form

Maxwell Electrodynamics

$$dF = 0$$

$$d\star F = 0$$

F is a two-form

Kalb-Ramond Electrodynamics

$$dH = 0$$

$$d\star H = 0$$

H is a three-form

Maxwell Electrodynamics

$$dF = 0$$

$$d\star F = 0$$

F is a two-form

Kalb-Ramond Electrodynamics

$$dH = 0$$

$$d\star H = 0$$

H is a three-form

Maxwell Electrodynamics

$$dF = 0$$

$$d\star F = 0$$

F is a two-form

Now proceed as you would with any other gauge theory!

$$dH = 0$$

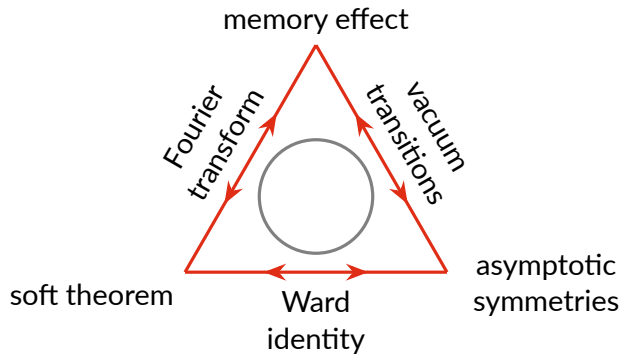
$$dH = 0$$

$$\therefore H = dB$$

$$dH = 0$$

$$\therefore H = dB$$

$$B \rightarrow B + dA$$



“Sound as a gauge theory and its infrared triangle”

Níckolas de Aguiar Alves & André G. S. Landulfo

arXiv: 2512.15796



See also:

Níck & Costa 2026, arXiv: 2609.20872

Níck 2026, arXiv: 2504.12521

